

Solve and Graph Inequalities

Lesson 8-6

Name:

Date:

Class:



TODAY'S OBJECTIVES

Content Objective: I can solve an inequality and graph its solution set on a number line.

Language Objective: I can explain my work using the words solve, graph, solution, and substitute.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Solve

Graph

Solution

Substitute

Solution set

Inverse operation



Tap any word to see its meaning and a picture.

Level 3: try the blanks from memory first, then check the bank.

1

To find all the values that make an inequality true is to

it.

2

To show the solutions on a number line is to them.

3

A value that makes an inequality true is a .

4

To check a solution, I the value back into the inequality.

5

All the values that make an inequality true form the

.

6 An operation that undoes another to isolate the variable is an

Write About the Math

How much more does the detective need? How did you solve and graph this inequality?

¿Cuánto más necesita la detective? ¿Cómo resolviste y graficaste esta desigualdad?

Say your answer to a partner first. Then write it, using at least two of these words:

☐ **Solve**
Resolver

☐ **Graph**
Graficar

☐ **Solution**
Solución

☐ **Substitute**
Sustituir

☐ **Solution set**
Conjunto solución

Model: *Solving an inequality gives a range of solutions, not one number.* — Students note both use inverse operations to isolate the variable, but an inequality's solution is many values (a range) while an equation's is a single value.

Start

Complete the frames. Use the word bank.

Completa las oraciones. Usa el banco de palabras.

- **The detective needs at least \$__ more because $48 + s \geq 75$ means $s \geq 75 - \underline{\hspace{1cm}} = \underline{\hspace{1cm}}$.**
- **My answer is __ because __.**

Mi respuesta es __ porque __.

Build

Write two connected sentences. Use because, so, or but.

Escribe dos oraciones conectadas. Usa porque, entonces o pero.

- **My claim is __.**
Mi afirmación es __.
- **My evidence is __, so __.**
Mi evidencia es __, entonces __.

Explain

Write a claim, give math evidence, and explain your reasoning.

Escribe una afirmación, da evidencia matemática y explica tu razonamiento.

☐ I answered the question.

☐ I used math evidence: numbers, an equation, or a model.

☐ I used at least two math words.